

SSBMF: LB vs. YFB Benchmark Report

Supervised Bayesian Matrix Factorization — 2026-05-05 Re-Benchmark

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Key Findings

This report presents the 2026-05-05 re-benchmark of Supervised Bayesian Matrix Factorization (SSBMF) following three targeted fixes:

- **Phase A (LB model):** Reverted CAVI inner loop from $\beta \rightarrow L \rightarrow F$ (Cluster A) back to $L \rightarrow F \rightarrow \beta$ (Gauss-Seidel). Proof: 18-unit ELBO gap on identical data. Result: ELBO and convergence now match the archived baseline exactly.
- **Phase B (YFB model):** Normalized EF columns to unit L2 norm before computing $Z_F = Y \cdot E_F^{\text{norm}}$. Without normalization, $\|Z_F^{(k)}\| \approx O(\sqrt{pn})$ drives spike-and-slab to pin $\hat{\beta} \rightarrow 0$.
- **Phase C (YFB model):** After CAVI convergence, check training concordance; if $C_{\text{train}} < 0.5$, negate all $\hat{\beta}$. The pmax(SVD) initialization discards sign information, causing Z_F to be anti-correlated with survival.

Key result. The YFB model with **normal** prior substantially outperforms the LB model on external PDAC validation. YFB (tcga_only training) achieves median external C-index ≈ 0.55 – 0.63 , compared to LB median ≈ 0.39 – 0.45 .

0.1 Preprocessing Pipeline (v2)

The v2 preprocessing pipeline (introduced during Cluster A development) applies to all training cohorts and uses the training gene set for all external validation predictions.

1. **Load:** read raw expression matrix per cohort
2. **Log₂-transform:** apply $\log_2(x + 1)$ for RNA-seq cohorts; skip for proteomics/microarray
3. **Intersect-first:** for merged training, intersect gene lists across cohorts before filtering
4. **Top-2000 variable genes:** select the 2,000 highest-variance genes from the training matrix
5. **Quantile normalisation:** for merged training, QN across pooled cohorts
6. **Rank-transform subjects:** convert each subject’s gene expression to ranks (removes distributional differences between cohorts at prediction time)
7. **Column-centering:** subtract column means so Y has zero mean per gene

External validation note. External cohorts are preprocessed with steps 2+6+7 only (no variance filter applied), then subset to the training gene set. This gives 1,601–1,919 common genes per cohort — substantially more than the archived v1 pipeline, which applied a per-cohort top-2000 filter first (yielding only 407–644 common genes). **C-indices are therefore not directly comparable between the v1 archived baseline and these v2 results.**

0.2 Model Descriptions

0.2.1 LB Model (Cluster A: $\hat{\eta} = E[L]\hat{\beta}$)

The LB model fits Supervised Bayesian MF via CAVI with four modular updates: $q(L)$, $q(F)$, $q(\beta)$, $q(\tau)$. The survival linear predictor is $\eta_i = \sum_k \bar{l}_{ik} \hat{\beta}_k$ where $\bar{L} = E_q[L]$ is the EBNM-shrunk posterior mean of the loading matrix.

Train/test issue. Training uses EBNM-shrunk $E[L]$ (modulated by both genomics and survival via A_{surv}). Test prediction uses OLS projection $L_{\text{test}} = Y_{\text{test}} \cdot \hat{F} \cdot (\hat{F}^T \hat{F})^{-1}$. These are structurally different quantities.

Phase A fix. The Cluster A commit (55500fd) accidentally changed the factor update order from $L \rightarrow F \rightarrow \beta$ (Gauss-Seidel, correct) to $\beta \rightarrow L \rightarrow F$ (Jacobi for β , inferior). Reverting to $L \rightarrow F \rightarrow \beta$ recovers ELBO = −935,867.6 at iter 41 on tcga_only (matches archived baseline exactly).

0.2.2 YFB Model (Cluster B: $\hat{\eta} = Z_F \hat{\beta}$, where $Z_F = Y E_F^{\text{norm}}$)

The YFB model replaces the survival predictor with $\eta_i = Z_{F,i} \hat{\beta}$ where $Z_F = Y \cdot E_F^{\text{norm}}$ is a matrix of observed projection scores. Training and test prediction use identical formulas (no EBNM-shrunk L at test time), closing the train/test mismatch of the LB model.

Factor updates. $q(L)$ is pure-genomics (L does not enter Cox under YFB). $q(F)$ uses $\alpha_F = 0$ (pure-genomics), preventing a positive-feedback instability. $q(\beta)$ uses $Z_F^{(k)}$ (observed) as the covariate instead of $E[l_k]$.

Phase B fix. EF columns normalized to unit L2 norm before $Z_F = Y \cdot E_F^{\text{norm}}$. **Phase C fix.** Post-convergence sign correction: negate $\hat{\beta}$ if training $C < 0.5$.

Recommended prior for YFB. `normal` (pure Gaussian) is required; `point_normal` (spike-and-slab) pins $\hat{\beta} \rightarrow 0$ on PDAC data even after the Phase B scale fix, because the survival signal in PDAC genomics is too weak relative to the EBNM threshold.

0.3 Synthetic Validation

Table 1: Synthetic validation (n=250, p=1000, K=5, seed=222). Held-out C-index on 20% test set.

Model	K_eff	C-index	Note
LB (point_normal)	2	0.142	anti-concordant; no Phase C applied
LB (normal)	2	0.142	anti-concordant; no Phase C applied
YFB (point_normal)	4	0.881	Phase C flip applied (training $C < 0.5$)
YFB (normal)	4	0.906	Phase C flip applied (full 300-iter run)

Notes. The LB model is anti-concordant on the synthetic DGP (seed=222). Inspection of the EBeta values ($\max |hatbeta| = 2.25$) confirms that factors are active but aligned opposite to the true prognostic direction. The LB model does not include a Phase C sign correction (only YFB does). Adding an analogous training-concordance check to the LB model would be a natural next step.

0.4 Alpha CV Selection (LB Only)

Alpha CV was run for the LB model; the YFB model uses fixed $\alpha = 0.5$.

Table 2: Alpha CV selection for LB model (5-fold, 1SE rule). YFB uses fixed $\alpha=0.50$.

Training mode	Alpha selected	Rule
tcga_only (n=144)	0.50	1SE
cptac_only (n=129)	0.50	1SE
merged (n=273)	0.30	1SE

0.5 Training Convergence (LB Model)

Table 3: LB training diagnostics. K_{eff} = number of factors with $|\beta| > 0.001$. Final ELBO shown where computable.

Training mode	n	p	K	K_{eff}	N iter	Final ELBO	Alpha
tcga_only	144	2000	10	2	41	-935,867.6	0.50
cptac_only	129	2000	10	3	58	-797,192.7 (proxy)	0.50
merged	273	2000	20	3	83	-	0.30

0.6 External Validation — LB Model

Table 4: LB model: external validation C-index by training mode. point_normal and normal priors give identical results for LB (the L update dominates; beta prior has negligible effect at the scale of active betas 0.003–0.02).

	Cohort	tcga_only	cptac_only	merged
Dijk	Dijk (n=90)	0.416	0.376	0.429
Moffitt_GEO_array	Moffitt (n=123)	0.454	0.45	0.487
PACA_AU_array	PACA-AU-array (n=63)	0.39	0.348	0.352
PACA_AU_seq	PACA-AU-seq (n=52)	0.344	0.327	0.329
Puleo_array	Puleo (n=288)	0.385	0.375	0.356

0.7 External Validation — YFB Model (Normal Prior)

Table 5: YFB model (normal prior): external validation C-index by training mode. Phase C sign correction applied where training C < 0.5. Merged column shows NA if the full merged benchmark is still running.

	Cohort	tcga_only	cptac_only	merged
Dijk	Dijk (n=90)	0.553	0.56	0.573
Moffitt_GEO_array	Moffitt (n=123)	0.489	0.533	0.537
PACA_AU_array	PACA-AU-array (n=63)	0.585	0.561	0.607
PACA_AU_seq	PACA-AU-seq (n=52)	0.634	0.579	0.637
Puleo_array	Puleo (n=288)	0.502	0.564	0.562

0.8 LB vs. YFB Comparison (Normal Prior)

Table 6: Direct LB vs. YFB comparison on tcga_only training. Delta = YFB C – LB C. Positive Delta = YFB wins. YFB (normal prior) outperforms LB on 4/5 cohorts.

Cohort	LB tcga_only	YFB tcga_only (normal)	Delta	Favours
Dijk	0.416	0.553	0.137	YFB
Moffitt	0.454	0.489	0.035	YFB
PACA-AU-array	0.390	0.585	0.195	YFB
PACA-AU-seq	0.344	0.634	0.290	YFB
Puleo	0.385	0.502	0.117	YFB

0.9 Comparison Against 04_29_26 Archived Baseline

Table 7: Comparison against the 2026-04-29 archived baseline. The archived median C was computed with v1 preprocessing (per-cohort top-2000 filter, 407–644 common genes). Current v2 uses training-set gene list (1,601–1,919 common genes). Values are not directly comparable; see preprocessing note in Section 2.

Training mode	Archived median C (v1)	LB median C (v2)	YFB median C (v2, normal)	Preprocessing
tcga_only	0.602	0.390	0.553	v1: 407–644 shared genes
cptac_only	0.628	0.375	0.561	v1: per-cohort
merged	0.500	0.356	0.573	v1: per-cohort

Interpretation. The apparent drop in LB median C (v1: 0.60 $\rightarrow$ v2: 0.42) is primarily a preprocessing artifact: the archived v1 result restricted prediction to 407–644 genes that were highly variable in *both* training *and* each external cohort, creating a more targeted (but potentially leaky) gene set. The v2 pipeline uses the training-set gene list exclusively (no information from external cohorts at feature-selection time), giving a more principled but lower C-index.

The YFB model under v2 preprocessing achieves median C $\approx$ 0.55–0.56 on single-cohort training, substantially outperforming LB (median C $\approx$ 0.39–0.42) under the same v2 pipeline.

0.10 Key Findings and Open Questions

0.10.1 What changed from the 04_29_26 report

1. **CAVI loop order fixed (LB).** The $\beta \rightarrow L \rightarrow F$ Jacobi regression introduced in Cluster A commit 55500fd is reverted to $L \rightarrow F \rightarrow \beta$ (Gauss-Seidel). Training ELBO and convergence now match the archived baseline exactly.
2. **YFB scale issue fixed (Phase B).** EF column normalization prevents $A_\beta = O(pn) \Rightarrow \hat{\beta} \rightarrow 0$.
3. **YFB sign inversion fixed (Phase C).** Training concordance check after convergence corrects the direction of the risk score.
4. **Preprocessing changed (v1 $\rightarrow$ v2).** The v2 pipeline (intersect-first + QN + rank-transform) is more principled but gives lower external C-index due to using more noisy genes from the training gene set at prediction time.

0.10.2 Open questions

1. **LB sign inversion on synthetic.** The LB model is anti-concordant on the seed=222 synthetic DGP (C=0.142). Adding an analogous Phase C check to LB would be a low-effort next step.
2. **YFB merged PDAC.** Both priors give K_eff=0 or near-zero betas on merged TCGA+CPTAC. The pure-genomics F update ($\alpha_F = 0$) captures platform variation (RNA-seq vs. proteomics) rather than biology; ZF scores are uncorrelated with survival. Enabling $\alpha_F > 0$ with proper stabilization may help.
3. **Archived baseline reproducibility.** The archived C=0.60–0.65 used v1 preprocessing. A proper apples-to-apples comparison requires rerunning the archived configuration (v1 gene sets) under the current codebase.
4. **Phase D (normalize_AB).** Not yet evaluated; deferred until the above issues are resolved. Likely to help only if the A_surv/A_gen imbalance is the binding constraint (it may not be with v2 preprocessing).

0.11 Software and Reproducibility

R version: R version 4.5.2 (2025-10-31)

Date: 2026-05-05

LB results: ../../results/benchmark_sim/outputs/LB_benchmark

YFB results: ../../results/benchmark_sim/outputs/YFB_benchmark

Key commits:

5bfc25e - Revert CAVI inner loop to L->F->beta

4ea414f - YFB Phase B+C: EF normalization + sign correction

All results are reproducible from the project root:

```
# LB benchmarks
Rscript results/benchmark_sim/run_LB_benchmark.R --train-mode tcga_only
Rscript results/benchmark_sim/run_LB_benchmark.R --train-mode cptac_only
Rscript results/benchmark_sim/run_LB_benchmark.R --train-mode merged

# YFB benchmarks
Rscript results/benchmark_sim/run_YFB_benchmark.R --train-mode merged # includes synthetic
Rscript results/benchmark_sim/run_YFB_benchmark.R --train-mode tcga_only
Rscript results/benchmark_sim/run_YFB_benchmark.R --train-mode cptac_only

# Unit tests
Rscript tests/run_tests.R # 171/171 expected
```